

nuclear energy needed a device to initiate and control a sustained nuclear chain reaction and so we have the Nuclear Reactor. Four distinct 'Generations' of nuclear plants have existed through time.

Generation 1 reactors were the first to produce civilian nuclear power. Generation 2 designs were commercial reactors, but had the disadvantage of needing manpower to operate and not functioning if the power was shut off. This brings us to Generation 3 reactors which can function by themselves hence addressing, to an extent, safety concerns of working in a nuclear plant. And finally, Generation 4 reactors which right now are still in theory but will supposedly surpass all other nuclear reactors.

Obninsk Nuclear Power Station, situated 110 km from Moscow, was the first civilian nuclear power station in the world.

Operating Systems

"Real programmers program in binary" So goes a famous joke in the programming world, especially popular with the performance nazis. Truth be told, programmers programmed in binary once but this was when computers were just in their infancy. You'd punch your instructions on a card in the form of dots representing the binary instructions and feed them to the machine; the machine would then execute them. Any errors were only to be blamed on you. Since then, computing has come a long way, thanks to operating systems (OSes).

An OS forms a layer between the hardware and the application program while managing other programs and keeping an account of hardware resource usage. The program asks the OS for the resources (network, disk, display etc.) and doesn't have to deal with the hardware level details such as disk configurations, display resolution etc. In early days, OSes were written in the assembly language. These included OSes such as MULTICS and the first version of UNIX which were written in the assembly language of their target processor architectures. However, later Ken Thompson (also the creator of language 'B') and Dennis Ritchie created the C programming language to port UNIX to PDP-11 and since then, OSes have been written in C.

The invention of the OS started the true software revolution by reducing the amount of effort required to create a working piece of code. This encouraged more and more geeks to try out programming and discover new capabilities. To this day, almost any device that can handle number crunching runs some form of an OS.